Oli Pratt | Engineering portfolio

Motorsport Engineering student · University of Derby

I'm a second-year BEng (Hons) Motorsport Engineering student looking for a 2027–28 industrial placement. I'm particularly interested in race engineering, vehicle performance and how teams use data to make decisions.

Projects

Brands Hatch driver comparison | MoTeC i2 Pro

I compared two drivers' laps at Brands Hatch using race data supplied for a university exercise. Looking at the plotted driving paths alongside in-car video, I found that the quicker driver took a noticeably tighter line through Turn 2.

Component modelling | SolidWorks

I built a 3D part from a supplied 2D drawing in an individual CAD exam and produced a dimensioned drawing from my model. I scored 84% overall, with full marks for model accuracy.

Numerical programming | MATLAB

I wrote and ran my own MATLAB code using vectors, matrices, functions and loops, then presented the results in tables and labelled plots. I checked a loop calculation against MATLAB's built-in sum function. I scored 94% for the coursework.

Gas-powered model car | Workshop manufacture

I laser-cut plywood parts, waterjet-cut and bent sheet metal, injection-moulded the wheels and cut and finished the axles by hand. I assembled the car and it ran successfully in an informal test.

Practical engineering

Vehicle measurement and testing

Working with other students, I measured corner weights, wheelbase and tyre rollout on a Radical SR1 for a centre-of-gravity exercise. I also completed practical sessions in engine component measurement, wind-tunnel testing and small-engine dynamometer testing.

ECU and data logging

I configured a start-up tune in TunerStudio and tested it on an ECU bench board. I also worked on logger and CAN configuration in Cosworth Toolset; our lecturer tested the dashboard configuration on the university's Formula Student car.

Education and training

University of Derby: BEng (Hons) Motorsport Engineering, Year 2.

Completed MATLAB Onramp and the MySolidWorks CSWA Exam Prep Course. Simulink Onramp is in progress, and I plan to sit the CSWA exam in January 2027.

St Albans School: A levels in Mathematics, Further Mathematics and Physics. I also studied part of the Computer Science A-level course, covering Python, data theory and HTML. I achieved grade 9 at GCSE in Computer Science, Mathematics, Physics and Biology.

Other experience

I've tutored GCSE maths and science one-to-one, completed student representative training at Derby and worked as a delivery driver alongside university.

Placement enquiries: oliver@oliverpratt.co.uk